

CODE FOR HARDWARE

```
#include <WiFi.h>

#include <LiquidCrystal_I2C.h>

// WiFi Credentials

const char* ssid = "YOUR_SSID";

const char* password = "YOUR_PASSWORD";

// Flow Sensor Pins

#define FLOW1_PIN 4

#define FLOW2_PIN 5

volatile int pulseCount1 = 0;

volatile int pulseCount2 = 0;

float flowRate1 = 0.0;

float flowRate2 = 0.0;

// LCD I2C (0x27 or 0x3F)

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Interrupt Functions

void IRAM_ATTR pulseCounter1() {
```

```
pulseCount1++;

}

void IRAM_ATTR pulseCounter2() {

    pulseCount2++;

}

void setup() {

    Serial.begin(115200);

    pinMode(FLOW1_PIN, INPUT);

    pinMode(FLOW2_PIN, INPUT);

    attachInterrupt(digitalPinToInterrupt(FLOW1_PIN),    pulseCounter1,
FALLING);

    attachInterrupt(digitalPinToInterrupt(FLOW2_PIN),    pulseCounter2,
FALLING);

    lcd.init();

    lcd.backlight();

    // WiFi Connection

    WiFi.begin(ssid, password);

    lcd.setCursor(0,0);
```

```
lcd.print("Connecting WiFi");

while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

}

lcd.clear();

lcd.print("WiFi Connected");

delay(2000);

}

void loop() {

    pulseCount1 = 0;

    pulseCount2 = 0;

    delay(1000); // 1 second measurement

    // Calculate flow rate (example formula)

    flowRate1 = pulseCount1 / 7.5;

    flowRate2 = pulseCount2 / 7.5;

    lcd.clear();
```

```
lcd.setCursor(0,0);

lcd.print("F1:");

lcd.print(flowRate1);

lcd.print(" F2:");

lcd.print(flowRate2);

lcd.setCursor(0,1);

// Leakage Detection Logic

if (abs(flowRate1 - flowRate2) > 1.0) {

    lcd.print("LEAK DETECTED");

    Serial.println("Leakage Detected!");

    // Send to IoT (Example Serial)

    Serial.print("Flow1: ");

    Serial.print(flowRate1);

    Serial.print(" Flow2: ");

    Serial.println(flowRate2);

}

else {

    lcd.print("NORMAL FLOW");
```

```
Serial.println("Normal Condition");
```

```
}
```

```
delay(2000);
```

```
}
```